Solar and battery can reduce energy costs and provide affordable outage backup for U.S. households

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Abstract

Distributed energy resources are promising solutions for household energy affordability and resilience as weather extremes and aging infrastructure intensify grid reliability risks. This study presents a comprehensive nationwide assessment of over 500,000 U.S. households, evaluating economic and backup viability of solar-battery systems. We find that 60% of households could reduce electricity costs with average savings of 15%, while 63% of households could achieve affordable backup power during power outages covering an average of 51% of their essential energy needs. However, these benefits show limited alignment with areas of greatest need, particularly in regions facing high outage risks. We also identify significant disparities in access to solar and battery, with less-populated and disadvantaged communities showing consistently lower viability. These findings demonstrate the need for targeted policy interventions to ensure equitable access to solar-battery benefits, especially as states transition from net energy metering to other electricity tariff policies.

Main

Electricity is fundamental to modern life, powering essential services from lighting, cooling and heating in the built environment to manufacturing, healthcare, communication, transportation and data centers[1]. Its reliability and affordability directly impact human health, safety, and daily activities[2, 3]. However, two mounting challenges threaten energy security across the United States: aging infrastructure and the increasing frequency, intensity, and duration of climate change-driven extreme weather events[4]. These challenges not only escalate energy costs but also lead to more frequent and severe power outages, particularly endangering vulnerable populations during extreme weather conditions[5, 6].

For individual households, distributed energy resources (DERs), specifically solar photovoltaic (PV) and battery storage systems, present promising solutions through their decentralized nature and broad geographical reach[7, 8]. These technologies can transform how families power their homes by delivering clean energy and providing backup power during outages all while potentially reducing overall electricity costs—a win-win that makes them accessible to a broader population[9–11]. Yet, realizing this potential across diverse communities and regions presents a significant challenge contingent on policy and local conditions.

The policy landscape for DERs is evolving rapidly, with decisions being made today that will shape households' energy options for decades to come. Some U.S. states, including California, Arizona, and Indiana, are replacing net energy metering (NEM) with net billing tariffs (NBT) to address concerns over fairness, utility revenues, and battery storage adoption[12–14]. States covering 35% of the U.S. population have already moved to net billing, and more are debating similar shifts or imposing capacity caps on net metering (Supplementary Note 1). Net metering compensates households the retail rate for excess solar generation, whereas net billing compensates those exports at lower, wholesale-based values or avoided costs. This shift necessitates more nuanced DER viability assessments that account for diverse household electricity consumption patterns[15, 16] and emphasizes the role of battery storage, as net billing incentivizes combined solar-battery systems[14].

Previous research on household solar and battery access has focused on the economic feasibility for single or a few representative households across broad regions[17–19]. While these seminal studies demonstrated the financial viability of such systems, they lacked the granular detail needed to address specific where and who questions, limiting their policy relevance. Recent studies have expanded to consider the heterogeneity of household characteristics[20–25] and investigate system-resilience benefits during weather-related grid disruptions[24–27]. Some U.S.-focused analyses have evaluated detailed regional rooftop solar economics but often excluded battery storage[22, 23]. Others have examined solar-battery system resilience under various household consumption patterns but focused solely on technical feasibility while overlooking financial considerations[24, 25].

Despite growing interest, the viability of solar and battery systems for providing cost reduction and outage backup across diverse U.S. households and regions remains understudied. The alignment between system viability and areas of greatest need receives little attention, as do equity considerations in access to such systems.

While some studies have identified racial and ethnic disparities in existing solar deployment[28, 29], these analyses of historical installations might overlook regions with minimal adoption and do not capture emerging viability trends driven by rapidly evolving technology costs, electricity prices, and policy incentives.

To address these knowledge gaps, this study presents a high-resolution nationwide assessment of over 500,000 U.S. households' access to solar PV and battery storage, evaluating both cost reduction and affordable backup power provision potential. We introduce economic and backup viability metrics that provide spatially detailed insights for stakeholders at individual and regional levels. Our bottom-up approach reveals that while the majority of households could benefit from solar-battery installations, their potential for economic savings and outage backup sometimes misaligns with areas of greatest need. We also find that disadvantaged communities generally have worse access to these technologies, potentially exacerbating existing installation disparities. Lastly, we demonstrate that these metrics can inform policy decisions such as the ongoing NEM phase-out.

Solar-battery systems reduce household electricity costs

The economic viability of solar-battery systems is characterized by whether, and to what extent, a household can reduce its overall electricity costs through the installation of solar PV, battery storage, or both technologies in combination. A household is considered economically viable when its yearly electricity bill reduction—the difference between its electricity bill before and after installation—exceeds annualized capital and operational costs of the system. We conduct a comprehensive assessment of solar-battery system economic viability across extensive representative U.S. households (n=512,437), with a coverage of nearly all counties in the contiguous United States. This granular analysis incorporates regional variations in technology costs, electricity rates, solar insolation, and household electricity consumption patterns (see Methods).

Our economic viability analysis centers on scenarios where households optimize their solar-battery system sizing and operation to maximize electricity cost reductions, following what we term the household optimal economic plan. Here, a household's electricity cost reduction is computed as the difference between its electricity bill before installation and its electricity costs after installation, which include the annualized capital and operational costs in addition to the bill. We classify households for whom solar PV installation is included in their optimal economic plan as economically optimal solar households, while those for whom battery storage is included are designated economically optimal battery households. A household can qualify for both categories if the combined installation yields maximum cost reductions (see Methods).

Figure 1a shows that under NBT, a majority (60.3%) of households nationwide are economically viable. Nearly all economically viable households are also economically optimal solar households (60.0% of total), while 16.8% are economically optimal battery households, with 16.5% qualifying for both categories. The top ten states account for the majority of each category, driven especially by larger-population states like

California, Texas, Florida, and New York (Fig. 1b-1e). In contrast, the state-level proportion data (Fig. 1f) reveals that some smaller-population states (such as Nevada and Kansas) have notably high proportions of economically viable households even if they do not appear among the top ten by absolute numbers. County-level analyses (Fig. 2, The base shapefile for the map was obtained from the U.S. Census Bureau[30]) show pronounced differences across counties within the same state and further indicate variability among households in the same county (see Supplementary Note 2 for detailed state- and county-level analysis). In contrast, under NEM, more households are economically viable (81.1%) and economically optimal solar households (79.5%), yet substantially fewer (3.4%) are economically optimal battery households. This occurs because net metering allows households to export their surplus solar generation at retail rates, reducing the incentive to include battery storage in their household optimal economic plan. By comparison, net billing's lower export compensation rates increase the cost-effectiveness of battery for shifting excess generation to meet onsite demand, encouraging more self-consumption of solar energy (Supplementary Fig. 1).

While NBT enables electricity cost reductions for most households, the magnitude of these savings varies. To capture this variation, we use a complementary cumulative distribution (CCD) of household-level cost reductions (Fig. 3). The CCD shows, for any given cost reduction level r (in %), the proportion of households $P(r)$ achieving electricity cost reductions (in %) greater than r. The fraction of total electricity bills contributed by these households is $F(r)$. Formally, for the set of all households $\mathcal{N}$, where household i has cost reduction r_i and an electricity bill b_i: $F(r) = (\sum_{r_i > r} b_i)/(\sum_{i \in \mathcal{N}} b_i) \times 100\%$. The convex shape of the CCD curve suggests that most households achieve modest savings, with relatively few reaching higher reduction levels (only 11.5% of households surpass 25% savings). The shaded area under the CCD curve corresponds to the electricity bill-weighted average cost reduction across all households: $\bar{r} = \int_0^{r_{\max}} F(r)dr = (\sum_{i \in \mathcal{N}} b_i r_i)/(\sum_{i \in \mathcal{N}} b_i)$, which is 14.7%, with weighted standard deviation (wstd) being $\sqrt{(\sum_{i \in \mathcal{N}} b_i (r_i - \bar{r})^2)/(\sum_{i \in \mathcal{N}} b_i)}$=17.3%. In contrast, under NEM, the average cost reduction is 43.6% (wstd=25.8%; Supplementary Fig. 3), indicating the substantial impact of electricity tariff policies on potential savings. An extended analysis for both NEM and the non-export tariff (NET), a scenario where excess solar generation is not compensated, is provided in Supplementary Note 3.

Solar-battery systems provide affordable outage backup

By offsetting capital and operational costs through routine day-to-day bill savings, solar-battery systems can provide backup power during outages without imposing additional expenses on households. Backup viability refers to a household's ability to maintain affordable backup power using solar PV, battery storage, or both during grid outages. A household is considered backup viable if it can achieve any level of critical consumption backup—the fraction of essential energy needs (refrigeration, lighting, hot water, heating, cooling, and half of the plugged load) served during outages—without increasing overall electricity costs. As in the economic viability assessment, we evaluate backup viability nationwide for the same set of households, but we now also account for each household's potential outage scenarios and incorporate the value of lost load (VoLL)[31] to capture the economic impact of unserved energy (see Methods). Additionally, we assume that households installing solar PV without battery storage can still achieve solar-to-load capability during outages (e.g., through specialized inverters).

Our backup viability assessment examines scenarios where each household follows its household optimal backup plan, which sizes the solar-battery system to maximize critical consumption backup while maintaining affordability (i.e., ensuring overall electricity costs do not increase; see Methods). In other words, each system is designed to provide as much backup as possible without raising electricity costs. Households whose optimal backup plan includes solar PV are referred to as backup optimal solar households, while those including battery storage are termed backup optimal battery households. As with the economic viability, a household can belong to both categories.

Figure 4a shows that under NBT, 62.7% of households are backup viable. Among these, 61% of all households are backup optimal solar households, 40.4% are backup optimal battery households, and 38.7% qualify as both. Compared to economic viability, backup optimal solar households show a similar geographic concentration, while backup optimal battery households are more broadly distributed beyond the top ten states (Fig. 4b–4e). This broader footprint is also apparent in the generally higher proportion of backup optimal battery households at the state and county level relative to their economically optimal counterparts (Fig. 4f, Extended Data Fig. 1; see Supplementary Note 4 for details). Moreover, because economically viable systems inherently provide some level of backup, all economically viable households are also backup viable, although their systems may offer less backup capacity than those sized specifically for outages (see Supplementary Note 5).

The CCD in Fig. 5, analogous to Fig. 3 for cost reductions, illustrates household-level backup capability under NBT. For any given critical consumption backup k (in %), it shows both the proportion of households $Q(k)$ achieving backup greater than k and their corresponding fraction of total essential energy needs $G(k)$ during outages. Formally, $G(k) = (\sum_{k_i > k} c_i)/(\sum_{i \in \mathcal{N}} c_i) \times 100\%$, where household i has critical consumption backup k_i and essential energy needs c_i during outages. The concave CCD shape indicates that a substantial portion of households achieve relatively high backup levels (42% of households surpass $k = 50\%$ critical consumption backup). The shaded area under the CCD curve corresponds to the essential energy needs-weighted average critical consumption backup across all households: $\bar{k} = \int_0^{k_{\max}} G(k)dk = (\sum_{i \in \mathcal{N}} c_i k_i)/(\sum_{i \in \mathcal{N}} c_i) = 50.5\%$, with weighted standard deviation (wstd) being $\sqrt{(\sum_{i \in \mathcal{N}} c_i (k_i - \bar{k})^2)/(\sum_{i \in \mathcal{N}} c_i)} = 41.2\%$. Although this is lower than the 71% (wstd=38.1%) observed under NEM (Supplementary Fig. 11), it demonstrates that while net billing significantly reduces financial incentives for solar-battery installations, these systems can still deliver meaningful backup power during outages.

Detailed backup viability analyses under NEM and NET are provided in Supplementary Note 6. In addition, we examine backup viability of solar-battery systems under alternative specifications—such as zero VoLL conditions, prioritizing cost reduction, whole-home backup, and permitting limited cost increases. These supplemental analyses, presented in Supplementary Note 7, offer further insights into the robustness and adaptability of solar-battery backup solutions across different policy and operational contexts.

Evaluating household access to solar-battery systems

Building on analysis of economic viability (cost reduction potential) and backup viability (affordable backup potential), we undertake an in-depth evaluation of households' access to solar-battery systems by assessing these two viabilities across different states. To understand where the benefits of these systems are most urgently needed, we also incorporate two household-level contextual factors: electricity burden, defined as the proportion of household income spent on electricity, and outage risk, the proportion of household annual electricity consumption lost under worst-case power outage conditions when no backup systems are in place (see Methods). Integrating these dimensions provides a holistic view of households' solar-battery access, specifically how these systems can enhance energy affordability and resilience across diverse regions and populations.

Our state-level evaluation under NBT refines the national assessment (Fig. 3, 5) by constructing separate CCDs of electricity cost reduction and critical consumption backup for each state (Fig. 6). These CCD curves—tracking households above specific cost reduction, r, or backup thresholds, k—reveal distinct patterns that vary by state. Cost reduction curves (bronze) typically feature steep initial drops with extended tails, while backup curves (green) show more gradual declines. We derive each state's average electricity cost reduction and critical consumption backup (shown as colored text in Fig. 6) from the shaded areas under their respective CCD curves. Each state's semicircles display its average electricity burden (left) and outage risk (right). By combining these four dimensions, we not only obtain a comprehensive view of households' solar-battery access across states, but also uncover notable disparities. For example, while both Texas and Tennessee rank in the top quartiles of electricity burden and outage risk, Texas households achieve average cost reductions of 9.5% (wstd=5.5%) and backup of 68.1% (wstd=34.6%), whereas Tennessee households see only 4.8% (wstd=13%) and 21.4% (wstd=21.4%), respectively (details in Supplementary Note 8).

Having established these state-level averages of electricity cost reduction, critical consumption backup, electricity burden, and outage risk (lower 48 states plus the District of Columbia, N=49), we examine Pearson correlations among the four measures to understand how well solar-battery benefits align with energy affordability and grid reliability challenges. The strong correlation between electricity cost reduction and critical consumption backup ($r(47) = 0.81$, $p < 0.001$) indicates that states achieving greater percentage cost reductions also tend to realize higher levels of affordable backup power. In contrast, a modest positive correlation between electricity burden and outage risk ($r(47) = 0.32$, $p = 0.02$) suggests that areas with higher electricity costs are somewhat more prone to power disruptions. Likewise, the moderate correlation between cost reduction and electricity burden ($r(47) = 0.48$, $p < 0.001$) implies that households with higher electricity bills may gain some level of cost relief. Most notably, the near-zero correlation between critical consumption backup and outage risk ($r(47) = -0.1$, $p = 0.50$) reveals a deep misalignment: states facing pronounced grid reliability challenges do not see correspondingly higher backup improvements. Such patterns remain generally consistent at finer spatial scales, including county and household levels, and are robust to Spearman correlation tests for monotonic relationships (see Supplementary Note 8).

These findings reveal that while solar-battery systems can provide households with economic and resilience benefits, these benefits often fail to align with the areas that need them most. The misalignment is more pronounced for backup viability than economic viability, highlighting the need for targeted policy interventions. For example, policies could incentivize paired solar and battery installations in high-outage-risk regions to enhance affordable power backup capabilities. As climate change induced weather extremes intensify the frequency and severity of power outages, ensuring affordable and reliable backup solutions becomes increasingly critical for at-risk households.

The patterns of alignment and mismatch between solar-battery system capabilities and areas of greatest need generally persist across NEM and NET scenarios (Supplementary Note 9). We also compare solar-battery backup with a traditional generator-based approach in Supplementary Note 10, finding that although fossil fuel-based generators can deliver near-complete backup for some households, solar-battery systems remain more widely affordable overall. Additionally, we explored a scenario in which solar-battery systems are sized to cover all essential energy needs during outages, irrespective of the affordability constraints from the solar-battery access evaluation (see Supplementary Note 11).

Distributional equity in solar-battery access

Having characterized households' solar-battery access in terms of economic viability (cost reduction) and backup viability (affordable backup) of the systems, we investigate the distributional equity of these benefits—that is, whether they are fairly shared across different household and community contexts or if certain groups face structural inequities. We focus on six key metrics: economically optimal solar households, economically optimal battery households, electricity cost reduction, backup optimal solar households, backup optimal battery households, and critical consumption backup. Using a sample (512,437 households) that is broadly representative of the US population in terms of income and housing stock, we employ household-level binary logistic regression and linear regression to examine how various household- and county-level characteristics correlate with these metrics (see Methods).

Examining these viability metrics under NBT using regression analysis (logistic regression and linear regression) reveals significant inequities at both household and community levels (Fig. 7; complete regression results in Supplementary Table 8). For example, at the household level, greater average daily demand and load profiles featuring substantial peak-time consumption (higher price-load profile correlation) are associated with increases in viability across all six solar-battery metrics (see Household

Characteristics in Fig. 7a, 7b). These consumption patterns, often rooted in socioeconomic and behavioral factors that can be difficult to alter (e.g., housing characteristics and installed appliances, scheduling of employment, cooking, laundry, childcare, eldercare, etc.) can reinforce household-level disparities. At the community level, increased solar-battery viability for all six metrics is associated with higher population density, greater solar resource availability (solar full load hours), lower socioeconomic disadvantage, and availability of time-of-use (ToU) rates (see County Characteristics in Fig. 7a, 7b). Taken together, these findings suggest that emerging patterns of solar-battery access may deepen existing inequities in renewable energy access, such as historically lower rates of solar adoption in disadvantaged communities[28, 32] (see Supplementary Note 12 for detailed statistical analyses). Similar equity challenges are also observed under NEM and NET scenarios (see Supplementary Note 13).

Economic and backup viability for electricity policymaking

Economic and backup viability metrics offer a quantitative foundation for informed policymaking related to solar-battery access. To illustrate their policy relevance, we evaluate how transitioning from net metering to net billing affects key metrics at the state level for a subset of states—examining proportions of economically optimal solar and battery households, average electricity cost reduction, proportions of backup optimal solar and battery households, and average critical consumption backup (Fig. 8; see Supplementary Note 14 for all other states and parallel NEM-to-NET evaluations). This analysis provides policymakers with data-driven insights as they consider adjusting tariff structures to better align with evolving energy goals and market conditions.

Our analysis reveals distinct regional patterns in how this transition influences solar-battery access. For example, California, Nevada, and Florida show only moderate declines in electricity cost reductions while maintaining or improving other key metrics (Fig. 8, top rows). Notably, these states also see substantial increases in economically optimal battery households, suggesting that net billing more effectively encourages integrating battery storage with solar PV than net metering[14]. In contrast, states such as Iowa, Idaho, and Washington experience sizable declines across multiple viability metrics after the transition (Fig. 8, bottom rows). These divergent outcomes emphasize that while phasing out net metering can proceed with minimal adverse effects in some regions, it may substantially diminish solar-battery access elsewhere—highlighting the importance of carefully considering regional contexts and policy goals when adjusting tariff structures. In addition to the household perspective, net billing also enables electric utilities to retain notably higher revenues than net metering, reflecting greater financial sustainability for utilities (see Supplementary Note 15).

Discussion

Our findings show that a majority of U.S. households stand to benefit from solar PV and battery storage under NBT, with 60.3% potentially reducing their electricity costs and 62.7% achieving affordable backup power. On average, households achieve a 14.7% reduction in electricity costs or cover approximately 50.5% of their essential energy needs during outages, underscoring the significant potential of solar-battery systems to enhance energy affordability and resilience for households. At the same time, we observe substantial geographic variation in both economic and backup viability of solar-battery systems. For economic viability, we find a moderate positive correlation with electricity burden, echoing earlier evidence that higher retail electricity rates enhance DER cost-effectiveness[33]. In contrast, backup viability remains poorly aligned with regions facing frequent and severe outages. This discrepancy suggests that households in areas prone to outages may struggle to secure cost-effective backup solutions through solar-battery systems—particularly concerning as climate change intensifies outage risks and heightens the demand for resilient energy solutions[34]. To address this mismatch, targeted policy interventions may be necessary. For example, in states such as Louisiana and Tennessee, where high outage risks coincide with low backup viability, preliminary estimates suggest that providing annual incentives averaging \$211 and \$176 per household, respectively, for installing solar-battery systems as backup power solutions—which could offset annual financing or leasing payments—could raise all households' critical consumption backup above the national average (see Supplementary Note 16 for methodology and additional state-level estimates).

This large-scale, granular assessment provides an integrated evaluation of where solar-battery systems are viable and where their benefits are most needed, systematically integrating insights from research on techno-economic feasibility of DERs[21], their resilience benefits[24], household energy burdens[2], and the grid reliability challenges posed by power outages[35]. By situating solar-battery systems within the broader context of energy affordability and grid reliability, this work lays a foundation for future research at the intersection of DER deployment, energy resilience, and social equity.

Our results highlight persistent inequities in solar-battery access, with disadvantaged communities and less-populated areas consistently experiencing lower economic and backup viability—patterns observed across NBT, NEM, and NET scenarios. These inequities align with and help explain historically lower solar uptake in disadvantaged communities[28, 32] and raise the risk of widening adoption gaps over time. Targeted incentives, financing mechanisms, and community-based deployment programs could ensure that these systems deliver equitable benefits, preventing further entrenchment of energy disparities.

Methodologically, our bottom-up approach leverages household-level heterogeneity to deliver nuanced insights that surpass traditional aggregate analyses[17]. Although this approach demands extensive data, computational resources, and modeling sophistication (see Methods), ongoing advances in metering infrastructure[36] and computing capabilities[37] will facilitate increasingly fine-grained and policy-relevant assessments.

The metrics we develop—economic and backup viability—have broad potential applicability, guiding not only tariff reforms but also informing various policy decisions, including subsidy design[38], low-interest loan programs[39], consumer energy literacy initiatives[40], and infrastructure planning efforts[41]. Such metrics could provide useful quantitative guidance to policymakers as they navigate evolving market conditions and climate challenges.

We acknowledge several limitations. This study focuses on economic and resilience benefits, excluding other potential advantages such as deferred grid investments[42], emissions reductions[43], or reduced social vulnerability to power outages[44]. We do not incorporate local incentives, demand response programs, or the potential influence of electric vehicles. Moreover, our evaluation centers on viability rather than observed adoption and relies on current conditions in technology costs, electricity rates, power outages, climate patterns, and electricity consumption behaviors—all of which are likely to evolve, potentially altering the feasibility and benefits of solar-battery systems in the future. Additionally, because only about 40% of households in our sample currently use electric heating, broader electrification (e.g., installing heat pumps) could affect solar-battery viability (see Supplementary Note 17).

Future research could analyze DER benefits not explored here, incorporate changing technologies and market conditions, and employ more computationally efficient modeling approaches to broaden the scale and complexity of analyses. Privacy-preserving frameworks may facilitate data-sharing and collaboration with utilities, while expanding the scope to include commercial and industrial sectors would provide a more comprehensive view of DER impacts. Additionally, future work could explore community-scale approaches, such as mobile energy storage[45] and coordinated resource control, to further enhance resilience beyond the household level. Together, these efforts could help shape more equitable, resilient, and sustainable energy systems.

Methods

Data

Household Data

The household data for this study were derived from the 2022.1 Release of ResStock[46], a comprehensive end-use load profile dataset for residential households. ResStock includes data from approximately 550,000 household energy models across over 3,000 U.S. counties, using weather conditions from both a typical meteorological year and an actual meteorological year (2018). Each building energy model provides a 1-year energy consumption profile in 15-minute intervals, categorized by end-use. The dataset also provides demographic information for each household, including annual income, floor area, number of stories, and the number of occupants. We estimated the roof area for each household by dividing the floor area by the number of stories (we assumed a ground-coverage ratio of 98% for single-family and mobile homes and 70% for multi-family[24]), which helped determine the potential for solar panel installation.

We focused on the output from the actual meteorological year and selected a subset of 512,437 households, from which we excluded those that had more than three stories, as these are typically unsuitable for solar-battery system installations. In addition, for county-level analyses, we merged counties with fewer than five households with the nearest county in the same state to preserve sufficient diversity in housing characteristics. In analyses related to power outages, we considered refrigeration, lighting, hot water, heating, cooling, and half of the plugged load as households' essential energy needs during outages, corresponding to the relevant end-use categories within the ResStock dataset. This definition of essential energy needs was established after examining similar definitions from various sources, including the US Department of Energy's definition of building energy resilience[47], surveys on customer appliance usage preferences during outages[48], literature with similar definitions of critical loads[49], and electrification design standards from microgrid solution providers[50]. Unless otherwise specified, we used the Baseline Stock Representation scenario from ResStock for all analyses in this article.

As ResStock is designed to reflect the distribution of real buildings and housing stock as accurately as possible based on the best available data[46], we assume that the conclusions drawn from this representative sample of half a million households can largely be generalized to the U.S. population. In our subset households, 40.2% have incomes below 80% of their county's median household income, 18.7% are between 80% and 120%, and 41.1% exceed 120%, indicating a broad representation of income levels that aligns with real-world distribution patterns.

Retail Electricity Rate Data

The retail electricity rates used in this study were obtained from the Utility Rate Database (URDB)[51], which catalogs rate structures from over 3,700 U.S. utilities. We focused on residential rates active as of 2023 and mapped each household to every applicable rate based on the county-level utility service territories, as identified in the EIA-861 survey[52]. From these applicable rates, each household was then randomly assigned one, prioritizing those with ToU structures to better reflect conditions that facilitate solar-battery integration. We did not consider low-income or other specialized rate programs that certain utilities may offer, which could influence solar-battery viability for eligible households. The binary indicator of ToU rate availability at the county level was also derived from EIA-861 data.

Energy Generated by Solar PV

We obtained the annual hourly electricity production profile of a 1 kW solar PV system for all counties in the contiguous United States from NREL's PVWatts calculator[53]. This tool estimates energy production from PV systems based on weather inputs and PV system characteristics through hourly simulations over a year. Weather inputs included direct normal radiation, diffuse horizontal radiation, global horizontal radiation, dry bulb temperature, wind direction, and wind speed, provided by the ResStock dataset at the county level. These weather inputs correspond to the same actual meteorological year (2018) used in the ResStock simulations, ensuring temporal consistency between household load profiles and solar generation profiles. The PV system characteristics were set to standard module type, an azimuth of 180 degrees, a fixed roof-mounted array type, a system loss of 14%, and an inverter efficiency of 96%.

The hourly solar generation profile for the 1 kW system was converted to 15-minute intervals by assuming constant solar irradiation within each hour. In the assessment of household-level solar-battery system viability, we scaled the 1 kW reference profile to the desired system capacity (in kW) by multiplying the capacity by the 1 kW generation profile for the corresponding county.

Costs and Specifications of Solar-Battery Systems

The specifications and cost data for solar-battery systems were sourced from the 2023 NREL benchmark reports[54]. For solar PV panels, we used a module area of 1.77 m^2, a PV panel density of 203 W/m^2, and a lifetime of 25 years. The cost structure of solar PV was simplified to a fixed cost of $1,628 for permitting, inspection, and interconnection (PII), a variable cost of $2,131.5/kW proportional to system capacity, and an operations and maintenance (O&M) cost of $28.78/kW/year. For battery storage, we applied a power ratio of 0.4 (2.5-hour duration) and a lifetime of 15 years. The battery storage cost structure was simplified to a fixed cost of $1,633 for PII, a variable cost of $856.24/kWh proportional to capacity, and an O&M cost of $20.8/kWh/year. These values represent national averages but were adjusted to reflect state-specific costs using state-level system cost data (see Supplementary Note 18 for details on these adjustments). Each household was mapped to the corresponding solar-battery system costs based on its state location. A 30% of federal tax credit is assumed to be applied to solar and battery investment.

County-level Power Reliability Metrics

We obtained monthly System Average Interruption Duration Index (SAIDI) and System Average Interruption Frequency Index (SAIFI) values for U.S. counties spanning six years (2017–2022). SAIDI measures the total duration of power outages that an average customer experiences in a month, while SAIFI indicates the average number of power outages a customer faces in the same period. These indices were derived from county-level aggregations of power outage data compiled by PowerOutage.us, combined with county-level customer counts from Find Energy[55, 56]. The resulting SAIDI and SAIFI values served as inputs to the Power Reliability Event Simulation Tool (PRESTO)[57], which we used to generate power outage scenarios for subsequent analyses of each household (described later in this section).

Demographic Data

We used demographic data from multiple sources for this work. County-level population density information was obtained from the American Community Survey 5-year estimate data from 2018 to 2022[58]. We also collected the county-level index of deep disadvantage, an index representing a holistic view of disadvantage using health indicators, poverty metrics, and social mobility data[59]. State-level average weekly working hours information was obtained from the U.S. Bureau of Labor Statistics[60].

Wholesale Electricity Price and Avoided Cost for NBT

We assume that NBT compensates electricity sent back to the grid at wholesale electricity prices or avoided costs. For households in counties served by a regional transmission organization (RTO), compensation rates are based on wholesale electricity prices, sourced from the U.S. EIA Wholesale Electricity Market Data[61]. For households in counties not served by an RTO, the compensation rates are determined by the state's avoided cost, which was estimated as the average fuel cost per kWh for that state, derived from the EIA-860 survey[62]. To ensure consistency across simulations and due to data availability, wholesale electricity prices and fuel costs were taken from the year 2021.

Household electricity costs

We evaluated household electricity costs using two distinct cost formulations: one for the household optimal economic plan and another for the household optimal backup plan. The key difference between these two is that the backup plan includes the VoLL to account for the economic impact of unmet energy needs during outages.

Household electricity costs under the household optimal economic plan consist of two main components: the electricity bill paid to the utility and the capital and O&M costs of solar-battery systems, if present. The annualized household electricity costs are represented by:

$$AEC_{\mathrm{econ}} = b_{\mathrm{econ}} + ATC_{\mathrm{PV}} + ATC_{\mathrm{bat}}, \tag{1}$$

where AEC_{econ} is the annualized electricity cost, b_{econ} is the annual electricity bill under the economic plan, ATC_{PV} is the annualized total cost related to solar PV including capital and O&M, and ATC_{bat} is the annualized total cost related to battery storage. In this formulation, we assume a financing or leasing arrangement for solar and battery installations. When annualizing the capital costs, we do not explicitly account for the time value of money. This can be interpreted in two ways: either we assume zero-interest financing, or we recognize that any interest costs may be offset by rising retail electricity rates which would increase the financial returns from solar-battery systems, thus negating the need to account for either factor. Historical trends support this rationale—for instance, from 2010 to 2023, U.S. residential electricity rates have increased by nearly 40%[63].

The electricity bill is calculated based on the household's electricity consumption, solar generation, battery charging/discharging profiles, and the applicable retail electricity rates. For a household with a solar-battery system, the annual electricity bill is given by:

$$b_{\mathrm{econ}} = f + \sum_{t \in \mathcal{T}} (p_t[D_t + S_t - V_t]^+ - q_t[V_t - D_t - S_t]^+), \tag{2}$$

where $\mathcal{T}$ is the set of all time periods in a year (comprising 35,040 elements, with each element corresponding to a 15-minute interval), f is the fixed annual fee from utilities, and for each $t \in \mathcal{T}$: D_t is the household's electricity consumption, S_t is the energy charged to (or discharged from, if negative) the battery, V_t is the energy generated by solar PV, p_t is the volumetric retail electricity rate, and q_t is the compensation rate for electricity sent back to the grid. The term $[D_t + S_t - V_t]^+$ represents the portion of demand met by the utility, while $[V_t - D_t - S_t]^+$ represents the electricity sent back to the grid. Here, the $[\cdot]^+$ operator returns the non-negative part of its argument.

The compensation rate q_t depends on the electricity tariff policy: it is 0 under NET, equal to wholesale electricity prices or avoided costs under NBT, and equal to p_t for NEM (see Data subsection within Methods for details on how wholesale electricity prices and avoided costs are determined). We assume that under both NEM and NBT, any incentives for excess energy exported to the grid apply equally to solar PV-generated power and power discharged from battery storage.

The solar generation profile V_t is determined by the solar PV capacity C and a county-level, weather-driven annual electricity production profile per unit capacity, v_t (see Data subsection within Methods for details on how v_t is derived):

$$V_t = C \cdot v_t, \tag{3}$$

We simplify the cost structure for solar PV and battery storage into fixed costs, variable costs proportional to capacity, and O&M costs (See Data subsection within Methods for full cost specifications). The annualized total cost related to solar PV is given by:

$$ATC_{\mathrm{PV}} = IS \cdot F_{\mathrm{PV}} + V_{\mathrm{PV}}C + M_{\mathrm{PV}}C, \tag{4}$$

where IS is the binary indicator for the presence of solar PV, F_{PV} is the annualized fixed cost for PII in \$/year, V_{PV} is the annualized variable cost in \$/kW/year, and M_{PV} is the annual O&M cost per unit capacity in \$/kW/year. The annualized total cost associated with battery storage is given by:

$$ATC_{\mathrm{bat}} = IB \cdot F_{\mathrm{bat}} + V_{\mathrm{bat}}E + M_{\mathrm{bat}}E, \tag{5}$$

where IB is the binary indicator for the presence of battery storage, F_{bat} is the annualized fixed PII cost for battery storage in \$/year, E is the battery's energy capacity, V_{bat} is the annualized variable cost in \$/kWh/year, and M_{bat} is the annual O&M cost per unit capacity in \$/kWh/year.

Under the household optimal backup plan, we incorporate not only the electricity bill and the solar-battery capital and operating costs (as in the economic plan) but also an annualized outage cost associated with unserved energy during power outages. Thus, the annualized household electricity costs under the backup plan are:

$$AEC_{\mathrm{backup}} = b_{\mathrm{backup}} + ATC_{\mathrm{PV}} + ATC_{\mathrm{bat}} + AOC, \tag{6}$$

where AEC_{backup} and B_{backup} represent the annualized electricity costs and the annual electricity bill under the backup plan, respectively, and AOC is the annualized outage cost. The annual electricity bill in the backup plan is:

$$b_{\mathrm{backup}} = f + \sum_{t \in \mathcal{T} \setminus \mathcal{O}} (p_t[D_t + S_t - V_t]^+ - q_t[V_t - D_t - S_t]^+), \tag{7}$$

where $\mathcal{O}$ is a set of time periods during which outages occur and $\mathcal{T} \setminus \mathcal{O}$ represents the set of time periods with no outage in the year.

The annualized outage cost is defined as:

$$AOC = \sum_{t \in \mathcal{O}} [D_t + S_t - V_t]^+ \times \text{VoLL}, \tag{8}$$

where $[D_t + S_t - V_t]^+$ represents the household's unmet electricity demand during an outage at time t, and VoLL quantifies the economic impact of unserved energy. The VoLL is estimated [64] based on state-level weekly working hours and the household's annual income (see Data subsection within Methods for details).

Generating power outage scenarios

We define a household's annual power outage scenario as a set of time periods, $\mathcal{O}$, encompassing every instance of power disruption within that year. To both explore long-term patterns and account for the uncertainty in future outages, we simulate 25 years (lifetime of solar PV) of potential outage scenarios for each household, drawing on historical outage data from the county in which the household is located. This process proceeds as follows:

First, we generate potential monthly power reliability metrics (SAIDI and SAIFI) for 25 years in each county. We accomplish this by randomly selecting one year of monthly SAIDI and SAIFI data from a six-year dataset (see Data subsection within Methods for details), then repeating this selection 25 times. For each household, we input the sampled one-year set of SAIDI and SAIFI values into PRESTO[57], a simulation tool that models the timing and duration of outages at the customer level based on county-level reliability metrics.

This procedure produces a single one-year outage scenario. By repeating it 25 times, we create a 25-year sequence of outage scenarios for each household. PRESTO is trained using PowerOutage.us data, ensuring consistency with our input metrics. Although we only have six years of county-level reliability data, reusing the same SAIDI and SAIFI inputs for multiple households within the same county—or even for the same household across different years—still introduces natural variability. This reflects the fact that even households in the same county can experience distinct outage events.

We denote the 25 distinct annual outage scenarios generated for each household as $\mathcal{O}_j$, where $j = 1, 2, ..., 25$.

Electricity burden and outage risk

Electricity burden and outage risk serve as contextual factors that indicate where reductions in electricity costs and the provision of backup power are most urgently needed.

We define the household's electricity burden as the percentage of its annual income spent on electricity before the installation of any solar or battery systems:

$$EB = \frac{AEC_{\text{econ}} \mid_{C=0,E=0}}{\text{Income}} \times 100\%. \tag{9}$$

Here EB is the electricity burden, and $AEC_{\text{econ}} \mid_{C=0,E=0}$ represents the household's annualized electricity costs in the absence of solar PV ($C = 0$) and battery storage ($E = 0$).

To quantify outage risk, we focus on the worst-case percentage of a household's electricity consumption that is unserved during power outages. Formally:

$$OR = \max_{j=1,\ldots,25} \left[\frac{\sum_{t\in\mathcal{O}_j} D_t}{\sum_{t\in\mathcal{T}} D_t} \times 100\% \right], \tag{10}$$

where OR denotes the outage risk and $\mathcal{O}_j$ is the set of outage periods in the j-th scenario. This measure identifies the scenario in which the household experiences the greatest fraction of unmet energy needs when no solar or battery systems are present.

Electricity cost reduction and critical consumption backup

Electricity cost reduction (in %) is defined as the percentage decrease in a household's electricity costs after installing solar PV and/or battery storage, compared to its pre-installation costs. For a given household, the electricity cost reduction is:

$$r = \frac{AEC_{\text{econ}} \mid_{C=0,E=0} - AEC_{\text{econ}}}{AEC_{\text{econ}} \mid_{C=0,E=0}} \times 100\%, \tag{11}$$

where r denotes the electricity cost reduction (in %). In this study, we calculate r assuming each household follows its optimal economic plan (introduced later).

Critical consumption backup is defined as the percentage of essential energy needs met during power outages. For a given household, the critical consumption backup over a year is:

$$k = \frac{\sum_{t\in\mathcal{O}} \min(V_t - S_t, \hat{D}_t)}{\sum_{t\in\mathcal{O}} \hat{D}_t} \times 100\%, \tag{12}$$

where k is the critical consumption backup (in %), and $\hat{D}_t$ is the household's essential energy needs in time period t. The min $(\cdot,\cdot)$ operation selects the smaller of the two values. Essential energy needs are defined as a subset of the household's total electricity consumption, including refrigeration, lighting, hot water, heating, cooling, and half of the plugged load. By definition, $\hat{D}_t \leq D_t$.

In the expression $\min(V_t - S_t, \hat{D}_t)$, $V_t - S_t$ represents the net available supply from solar and battery during an outage. It is always non-negative ($V_t - S_t \geq 0$) for $t \in \mathcal{O}$ since the battery cannot charge from the grid during outages. In this study, we report the value of k assuming each household follows its optimal backup plan (introduced later).

Economic viability and household optimal economic plan

To assess economic viability, we formulate each household's decision regarding solar PV and battery storage installations as a mixed-integer optimization problem. The objective is to minimize the household's annualized electricity costs (AEC_{econ}).

Mathematically, the household optimal economic plan is determined by solving the following optimization problem:

$$\text{Minimize} \quad AEC_{\text{econ}}. \tag{13}$$

The decision variables include a binary indicator for solar PV installation (IS), a binary indicator for battery storage installation (IB), solar PV capacity (C), battery storage's energy capacity (E), and the energy charged to or discharged from the battery at each time period (S_t).

Solar-battery system capacity limit constraints

$$IS \cdot C_{\min} \leq C \leq IS \cdot C_{\max}, \tag{14}$$

$$IB \cdot E_{\min} \leq E \leq IB \cdot E_{\max}. \tag{15}$$

Here, $C_{\min}$ and $C_{\max}$ denote the minimum and maximum permissible solar PV capacities, respectively. In our analysis, $C_{\min}$ is 0.72 kW (approximately two PV modules), and $C_{\max}$ is determined by each household's available roof area. Similarly, $E_{\min}$ and $E_{\max}$ represent the minimum and maximum allowable battery energy capacities, set at 3 kWh and 80 kWh, respectively[65]. This ensures that if a household installs solar or battery systems, the chosen capacities fall within realistic and policy-relevant bounds.

Battery operating constraints

The battery's charging and discharging behavior is governed by the following constraints, applied to each time period $t \in \mathcal{T}$:

$$-U\Delta t \leq S_t \leq U\Delta t, \, U = 0.4 \times E, \tag{16}$$

$$0 \leq H_t \leq E, \tag{17}$$

$$H_{t+1} = H_t + S_t, \, H_1 = 0.5 \times E. \tag{18}$$

Here, U is the battery's power capacity, defined with a power ratio of 0.4 relative to its energy capacity E. The variable Δt denotes the duration of each time period, and H_t represents the battery's state of charge (SoC) at the start of time period t. We assume that the initial SoC is half of the battery's energy capacity.

Nonnegative bill constraints

$$b_{\text{econ}} \geq 0. \tag{19}$$

This constraint prevents households from profiting through electricity exports (i.e., negative bills). While net-negative bills may be allowed in some regions, enabling households to earn profits from utilities, we do not consider this scenario in our study.

After solving the optimization, households with an optimal solution indicating $IS = 1$ (solar PV installed) are classified as economically optimal solar households, and those with $IB = 1$ (battery storage installed) are classified as economically optimal battery households. The achieved reduction in electricity costs, compared to the no-solar/no-battery baseline, defines the electricity cost reduction under the household optimal economic plan.

To manage computational complexity and streamline the analysis for half a million households, we adopt several simplifying assumptions. We treat solar PV and battery system capacities as continuous variables, rather than discrete market increments, allowing direct optimization without additional rounding steps. For battery performance, we assume a round-trip efficiency of 100% and no capacity degradation over its lifetime, thus enabling a one-year optimization horizon to approximate long-term behavior. Finally, we consider the chosen year of household electricity consumption and solar data as representative, assuming that the household's load and solar generation estimates remain sufficiently accurate for long-term evaluation. These approximations allow us to focus on the core economic dimensions of solar-battery system installation and operation, reducing computational demands while maintaining a realistic assessment.

Backup viability and household optimal backup plan

To assess backup viability, we define the household optimal backup plan in which the household sizes and operates its solar-battery system to maximize the fraction of essential energy needs served during power outages—referred to as critical consumption backup—without increasing overall electricity costs. Unlike the economic plan, where a single representative year of data is assumed, backup needs vary significantly from year to year. To capture this variability, we employ a Monte Carlo approach, generating multiple power outage scenarios spanning 25 years for each household. The household optimal backup plan is then determined by maximizing the expected critical consumption backup across these scenarios, subject to the constraint that expected household electricity costs do not exceed those without solar-battery installations.

Our analysis focuses on backup for electric loads, including electric heating where applicable. For households that rely on fossil fuels (e.g., natural gas, fuel oil, or propane) for heating, we assume continued access to these fuels during an outage if heating is necessary for habitability. We acknowledge that this assumption can understate the challenges faced in real events where fuel supply may also be disrupted. As a result, critical consumption backup described here applies exclusively to electricity loads, and may not capture the full extent of potential heating-related resilience issues in regions prone to simultaneous disruptions in both electric and fuel infrastructures.

We also account for the backup viability of households equipped only with solar PV and no battery storage. Although standard grid-tied solar systems shut down during outages for safety reasons, certain specialized technologies can enable solar-to-load capabilities. These solutions, such as Enphase Sunlight Backup[66] and SMA's Secure Power Supply[67], allow solar panels to power essential home circuits during daylight hours, even when disconnected from the grid. For our analysis, we assume that households with only solar PV have access to similar solar-to-load functionalities during outages.

Our objective is to maximize the essential energy needs-weighted average critical consumption backup ($\bar{k}$) across the empirical distribution of the 25-year outage scenarios:

$$\text{Maximize} \quad \bar{k}, \tag{20}$$

where:

$$\bar{k} = \frac{\sum_{j=1}^{25} \left[k \mid_{\mathcal{O}=\mathcal{O}_j} \cdot \sum_{t\in\mathcal{O}_j} \hat{D}_t \right]}{\sum_{j=1}^{25} \sum_{t\in\mathcal{O}_j} \hat{D}_t}. \tag{21}$$

Here, $k \mid_{\mathcal{O}=\mathcal{O}_j}$ is the critical consumption backup k achieved under outage scenario $\mathcal{O}_j$.

Similar to the household optimal economic plan, the decision variables include a binary indicator for solar PV existence (IS), a binary indicator for battery storage existence (IB), solar PV capacity (C), battery energy capacity (E), and the energy charged to or discharged from the battery ($S_t^{(j)}$) in each time period t under outage scenario $\mathcal{O}_j$.

Non-increasing household electricity costs constraints

To ensure affordability, we impose a constraint that the expected household electricity costs under the backup plan do not exceed those of a scenario with no solar or battery. This ensures that installing solar-battery systems for backup does not increase a household's overall electricity expenses relative to having no installations.

We define the empirical expectation $\mathbb{E}_{\mathcal{O}}[\cdot]$ over the 25-year set of generated power outage scenarios $\{\mathcal{O}_j\}_{j=1}^{25}$. If x denotes any quantity of interest (e.g., annual electricity costs) for scenario $\mathcal{O}_j$, then $\mathbb{E}_{\mathcal{O}}[x] = 1/25 \cdot \sum_{j=1}^{25} x \mid_{\mathcal{O}=\mathcal{O}_j}$. With this definition, the non-increasing cost constraint is:

$$\mathbb{E}_{\mathcal{O}}[AEC_{\text{backup}}] \leq \mathbb{E}_{\mathcal{O}}[AEC_{\text{backup}} \mid_{IS=0,\, IB=0}]. \tag{22}$$

Solar-battery system capacity limit constraints

These constraints mirror equations (14) and (15).

Battery operating constraints

For each power outage scenario $\mathcal{O}_j$ and each time period $t \in \mathcal{T}$, the battery's operation is governed by:

$$-U\Delta t \leq S_t^{(j)} \leq U\Delta t,\, U = 0.4 \times E, \tag{23}$$

$$0 \leq H_t^{(j)} \leq E, \tag{24}$$

$$H_{t+1}^{(j)} = H_t^{(j)} + S_t^{(j)},\, H_1^{(j)} = 0.5 \times E. \tag{25}$$

Here, $S_t^{(j)}$ represents the energy charged to (or discharged from, if negative) the battery at time t under outage scenario $\mathcal{O}_j$. The variable $H_t^{(j)}$ is the battery's SoC at the start of time period t under outage scenario $\mathcal{O}_j$.

Nonnegative bill constraints

Similar to the economic plan, we require the expected household electricity bill to be nonnegative under the backup plan. Taking the empirical expectation across the 25 power outage scenarios, we have:

$$\mathbb{E}_{\mathcal{O}}[b_{\text{backup}}] \geq 0. \tag{26}$$

Beyond the assumptions made for the household optimal economic plan, we also assume that household outage scenarios are known or can be reasonably estimated in advance. In practice, local utilities may issue planned outage notices related to maintenance, or reliable weather advisories may provide forewarnings of storms and extreme events, enabling households to prepare accordingly. However, some outages—such as sudden equipment failures—are inherently unpredictable, and households may receive no prior notice. Such unforeseen disruptions could reduce the actual backup viability compared to our optimized estimates. For computational efficiency, we further assume that the battery's SoC resets to 50% at the start of each week, simplifying the problem by decreasing the number of optimization variables.

Logistic and ordinary least squares regression models

We employ logistic regression for binary outcomes—such as whether a household is economically optimal solar/battery household or backup optimal solar/battery household—and linear regression using Ordinary Least Squares (OLS) for continuous outcomes, including electricity cost reduction and critical consumption backup. Through these models, we examine how various household- and county-level characteristics correlate with the observed solar-battery access metrics.

Household-level characteristics (see Data subsection within Methods for details) for household i are as follows: Annual income (Income_i), used in the regression as the natural logarithm of total annual household income (log($)); Average daily electricity demand (Demand_i), used as the natural logarithm of average daily household electricity consumption (log(kWh)); Number of occupants (Occupants_i), included in its original scale (integer count of people in the household); Solar–load profile correlation (SolarCorr_i), incorporated as the Pearson correlation coefficient (unitless) measuring how closely 15-minute solar generation and electricity load profiles align over the year; and Price–load profile correlation (PriceCorr_i), incorporated as the Pearson correlation coefficient (unitless) between 15-minute retail electricity prices and electricity load profiles.

County-level characteristics (see Data subsection within Methods for details) linked to household i are: Availability of ToU rates (ToU_i), a binary indicator (0/1) showing whether ToU rates are available in the household's county; Annual Average Customer Outage (Outage_i), included directly as the average annual customer duration of power outages (hours) in the county; Solar full-load hours (SolarHours_i), the theoretical number of hours a solar system would need to operate at its full rated capacity to produce its actual annual energy output, scaled so that a value of 1 represents 100 full-load hours of solar PV operation; Population density (PopDensity_i), used as the natural logarithm of the number of people per square mile (log(people/sq. mi.)); and Index of deep disadvantage [59] (Disadvantage_i), included in its original scale (-11 to 5), where higher values indicate more advantaged status (this index integrates health, poverty, and social mobility measures).

Logistic Regression Model

We use logistic regression models to estimate the likelihood that a household falls into specific categories—namely, economically optimal solar/battery households under the household optimal economic plan and backup optimal solar/battery households under the household optimal backup plan. Each logistic model estimates the probability $P(Y_i = 1)$ that a given household i belongs to one of these groups based on a set of household- and county-level characteristics.

The logistic model takes the form:

$$\begin{aligned}\text{Logit}(P(Y_i = 1)) &= \log\left(\frac{P(Y_i = 1)}{1 - P(Y_i = 1)}\right) \\ &= \beta_0 + \beta_1 \text{Income}_i + \beta_2 \text{Demand}_i + \beta_3 \text{Occupants}_i + \beta_4 \text{SolarCorr}_i \\ &+ \beta_5 \text{PriceCorr}_i + \gamma_1 \text{ToU}_i + \gamma_2 \text{Outage}_i + \gamma_3 \text{SolarHours}_i \\ &+ \gamma_4 \text{PopDensity}_i + \gamma_5 \text{Disadvantage}_i + \xi_i,\end{aligned} \tag{27}$$

where Y_i is a binary outcome indicating whether household i falls into the specified category. The coefficients β_m and γ_n represent the effects of household- and county-level variables on this probability, and ξ_i is the error term.

Linear Regression Model

For outcomes such as electricity cost reduction (under the economic plan) and expected critical consumption backup (under the backup plan), we employ linear regression. This model estimates a linear relationship between the outcome Z_i and the same household- and county-level characteristics used in the logistic models.

The linear regression model is expressed as:

$$\begin{aligned}Z_i = \alpha_0 &+ \alpha_1 \text{Income}_i + \alpha_2 \text{Demand}_i + \alpha_3 \text{Occupants}_i + \alpha_4 \text{SolarCorr}_i \\ &+ \alpha_5 \text{PriceCorr}_i + \delta_1 \text{ToU}_i + \delta_2 \text{Outage}_i + \delta_3 \text{SolarHours}_i \\ &+ \delta_4 \text{PopDensity}_i + \delta_5 \text{Disadvantage}_i + \epsilon_i,\end{aligned} \tag{28}$$

where Z_i represents the continuous outcome (e.g., electricity cost reduction or critical consumption backup), α_m and δ_n are the coefficients reflecting the influence of household- and county-level characteristics, and ϵ_i is the error term.

Computational methods

We implemented the optimization problems in this study using CVXPY[68] and solved them with Gurobi[69] and MOSEK[70]. Given the large number of households and scenarios, all simulations were executed on HPC clusters. See Supplementary Note 19 for details.

Data availability

The data used, generated and analyzed during this study are available through the Stanford Digital Repository at https://purl.stanford.edu/cn945xm8241 or https:

//doi.org/10.25740/cn945xm8241 (ref. [71]). Raw data sources (e.g., household energy consumption, building information, retail/wholesale electricity prices, solar PV generation, solar and battery capital and operating costs, power outage metrics, and social demographics) are either publicly available or purchased via data subscriptions; these sources are described in detail in the Data subsection of the Methods section.

Code availability

All code required to reproduce the results presented in this study is available through the Stanford Digital Repository at https://purl.stanford.edu/cn945xm8241 or https://doi.org/10.25740/cn945xm8241 (ref. [71]).

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T.S., C.Z., A.M., and R.R. conceived and designed the research; T.S. and Y.F. performed the experiments; T.S. and Y.F. analyzed the data; T.S., C.Z., and J.F.. contributed materials; T.S., Y.F., C.Z., J.F., A.M., and R.R. wrote the paper.

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Ethics declarations

Competing interests

The authors declare no competing interests.

Figure captions for main text figures

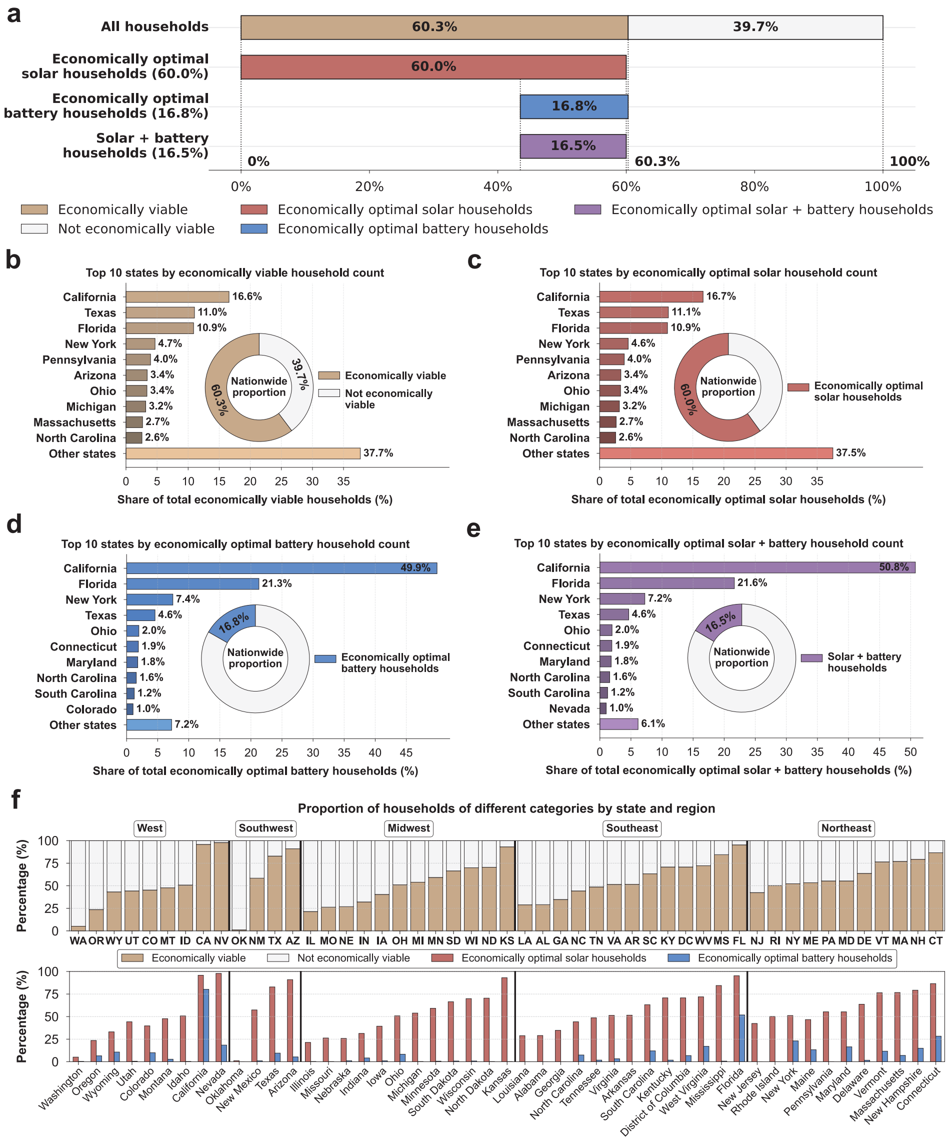

Fig. 1 Economically viable households under net billing tariffs. Households are economically viable if yearly bill savings from a solar-battery system exceed annualized system costs. A household is considered an economically optimal solar household if including solar photovoltaic (PV) maximizes cost reductions, an economically optimal battery household if including battery storage maximizes cost reductions, and an economically optimal solar + battery household if qualifying as both. **a,** Among 512,437 representative U.S. households, 60.3% are economically viable, 60% are economically optimal solar households, 16.8% are economically optimal battery households, and 16.5% qualify as both (solar+battery). **b–e,** National proportions (donut charts) of economically viable households (b), economically optimal solar households (c), economically optimal battery households (d), and households qualifying as both (e), along with each category's distribution in the top ten states (bar charts). **f,** State-level proportions of economically viable, economically optimal solar, and economically optimal battery households.

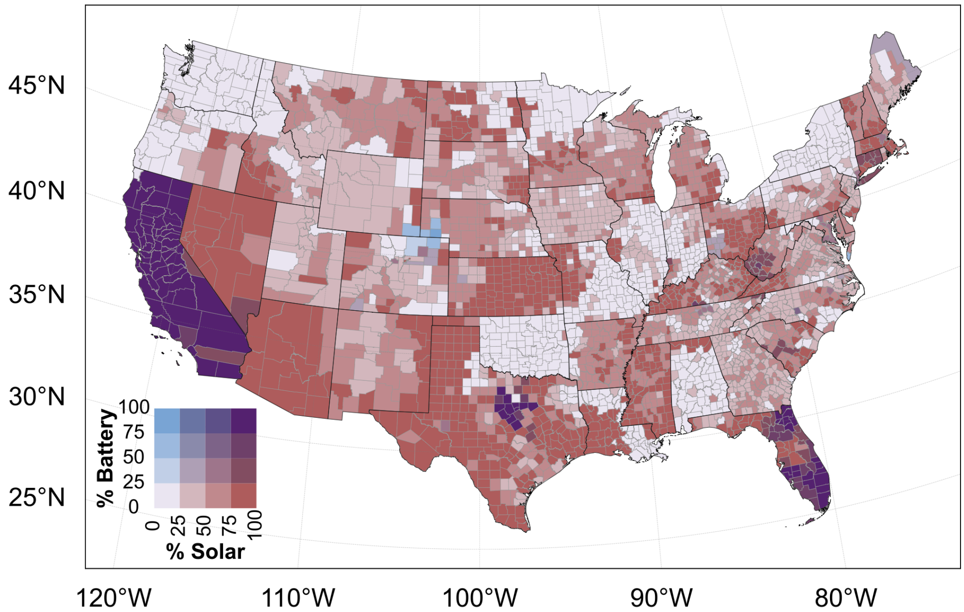

Fig. 2 County-level patterns of economically optimal solar and economically optimal battery households under net billing tariffs. The map displays two variables across counties in the contiguous United States: the proportion of economically optimal solar households (horizontal axis in the 4×4 color grid) and economically optimal battery households (vertical axis in the 4×4 color grid). Light purple indicates low proportions of both, while dark purple indicates high proportions of both. Red indicates a high proportion of economically optimal solar households with a low proportion of economically optimal battery households, whereas blue indicates the opposite.

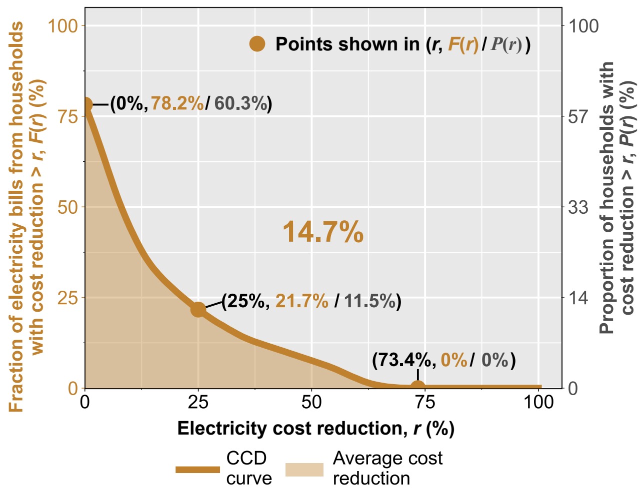

Fig. 3 Distribution of electricity cost reduction across households nationwide under net billing tariffs. Complementary cumulative distribution (CCD) of household-level electricity cost reductions (in %) for 512,437 representative U.S. households, where households size solar photovoltaic (PV) and/or battery storage to maximize their electricity cost reduction. For any cost reduction r (in %), the CCD shows both $P(r)$, the proportion of households achieving more than r cost reduction, and $F(r)$, the fraction of total baseline electricity bills contributed by those households prior to any installations. Formally, for the set of all households $\mathcal{N}$: $P(r) = n(r)/|\mathcal{N}| \times 100\%$, where $n(r)$ is the number of households exceeding r cost reduction, and $|\mathcal{N}|$ is the total number of households; $F(r) = (\sum_{r_i > r} b_i)/(\sum_{i \in \mathcal{N}} b_i) \times 100\%$, where r_i is the cost reduction of household i and b_i is its baseline electricity bill. At $r = 0\%$, 60.3% of households achieve positive cost reductions, representing 78.2% of total bills. The maximum observed cost reduction is $r_{\max}$=73.4%, at which point no households achieve greater savings. The shaded (bronze) area under the CCD curve represents the bill-weighted average cost reduction across all households, computed as $\bar{r} = \int_0^{r_{\max}} F(r)dr = (\sum_{i \in \mathcal{N}} b_i r_i)/(\sum_{i \in \mathcal{N}} b_i)$=14.7%, with weighted standard deviation being $\sqrt{(\sum_{i \in \mathcal{N}} b_i (r_i - \bar{r})^2)/(\sum_{i \in \mathcal{N}} b_i)}$=17.3%.

Fig. 4 Backup viable households under net billing tariffs. Households are backup viable if they can maintain a portion of essential energy needs during outages using solar-battery systems without raising electricity costs. Households whose optimal backup configuration includes solar photovoltaic (PV) are backup optimal solar households, those including battery are backup optimal battery households, and some qualify as both (solar+battery). **a,** Among 512,437 representative U.S. households, 62.7% are backup viable, 61% are backup optimal solar households, 40.4% are backup optimal battery households, and 38.7% for both. All economically viable households (60.3%) are backup viable; the 2.4% difference results from the value of lost load. **b–e,** National proportions of backup viable households (b), backup optimal solar households (c), backup optimal battery households (d), and households qualifying as both (e), with each category's distribution in the top ten states. **f,** State-level proportions of backup viable, backup optimal solar, and backup optimal battery households.

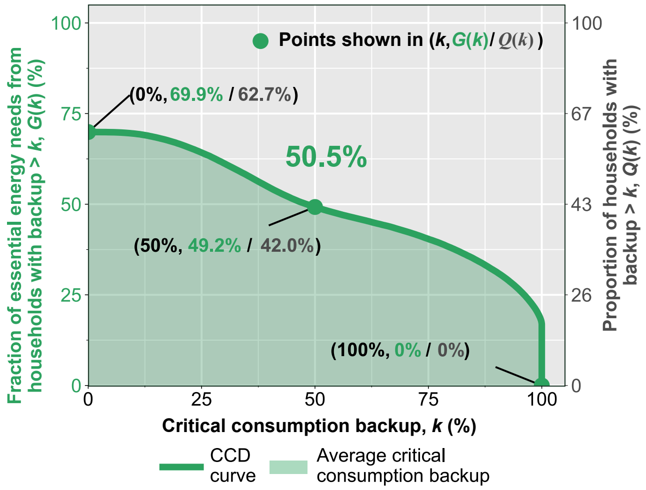

Fig. 5 Distribution of critical consumption backup across households nationwide under net billing tariffs. Complementary cumulative distribution (CCD) of household-level critical consumption backup, a measure of backup capability defined as the fraction of a household's essential energy needs that can be served during outages. Results are shown for 512,437 representative U.S. households which determine the optimal sizing of solar photovoltaic (PV) and/or battery storage to maximize this fraction without increasing overall electricity costs. For any given critical consumption backup k (in %), the CCD shows both $Q(k)$, the proportion of households achieving more than k backup, and $G(k)$, the fraction of total essential energy needs during outages accounted for by those households. Formally, $Q(k) = m(k)/|\mathcal{N}| \times 100\%$, where $m(k)$ is the number of households with $k_i > k$, and $G(k) = (\sum_{k_i > k} c_i)/(\sum_{i \in \mathcal{N}} c_i) \times 100\%$, where k_i is the critical consumption backup of household i, c_i is its essential energy needs during outages, and $\mathcal{N}$ is the set of all households. At $k = 0\%$, $Q(k) = 62.7\%$ of households achieve positive backup, representing $G(k) = 69.9\%$ of total essential energy needs. The maximum observed backup is $k_{max} = 100\%$. The shaded (green) area under the CCD curve equals the essential energy needs-weighted average critical consumption backup across all households, $\bar{k} = \int_0^{k_{\max}} G(k)dk = (\sum_{i \in \mathcal{N}} c_i k_i)/(\sum_{i \in \mathcal{N}} c_i)$=50.5%, with weighted standard deviation being $\sqrt{(\sum_{i \in \mathcal{N}} c_i (k_i - \bar{k})^2)/(\sum_{i \in \mathcal{N}} c_i)}$=41.2%.

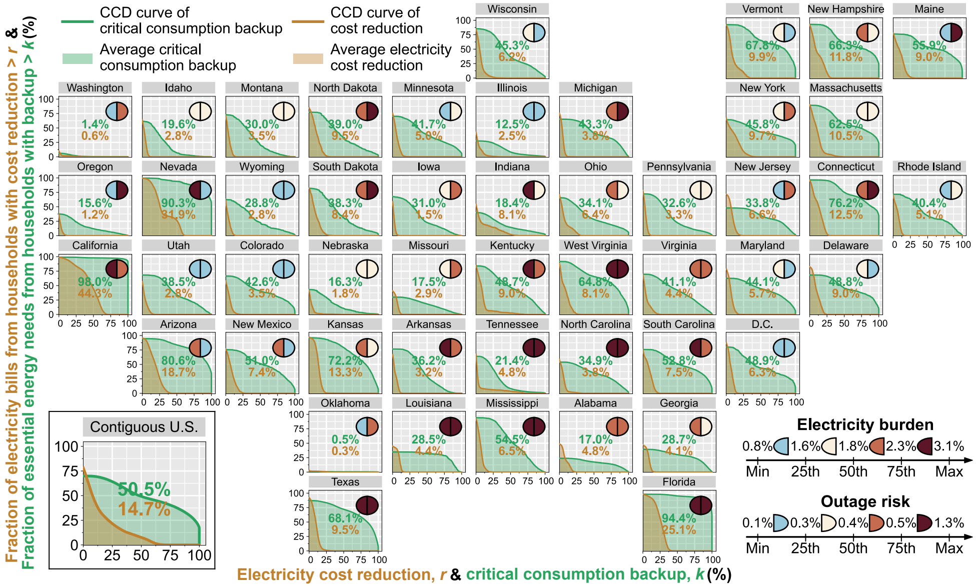

Fig. 6 State-level solar-battery access under net billing tariffs, contextualized by electricity burden and outage risk. By-state complementary cumulative distributions (CCDs) of household-level electricity cost reduction and critical consumption backup, combined with state-level quartile rankings of electricity burden and outage risk. The panel for the Contiguous U.S. combines the nationwide CCD curves shown separately in Fig. 3 (cost reduction) and Fig. 5 (backup) into a single display. The bronze CCD plots the fraction of households—weighted by their baseline electricity bills—achieving above each cost reduction threshold, while the green CCD plots the fraction of households—weighted by their essential energy needs during outages—surpassing each backup threshold. The shaded areas beneath these curves correspond to each state's bill-weighted average electricity cost reduction (bronze) and essential energy needs-weighted average critical consumption backup (green), displayed as colored text within each state's panel. Each state's electricity burden (fraction of household income spent on electricity) and outage risk (potential worst-case fraction of unserved energy during outages) are computed as income-weighted and consumption-weighted averages of household-level values, respectively, and represented by left (electricity burden) and right (outage risk) semicircles, with colors indicating their quartile rankings among states.

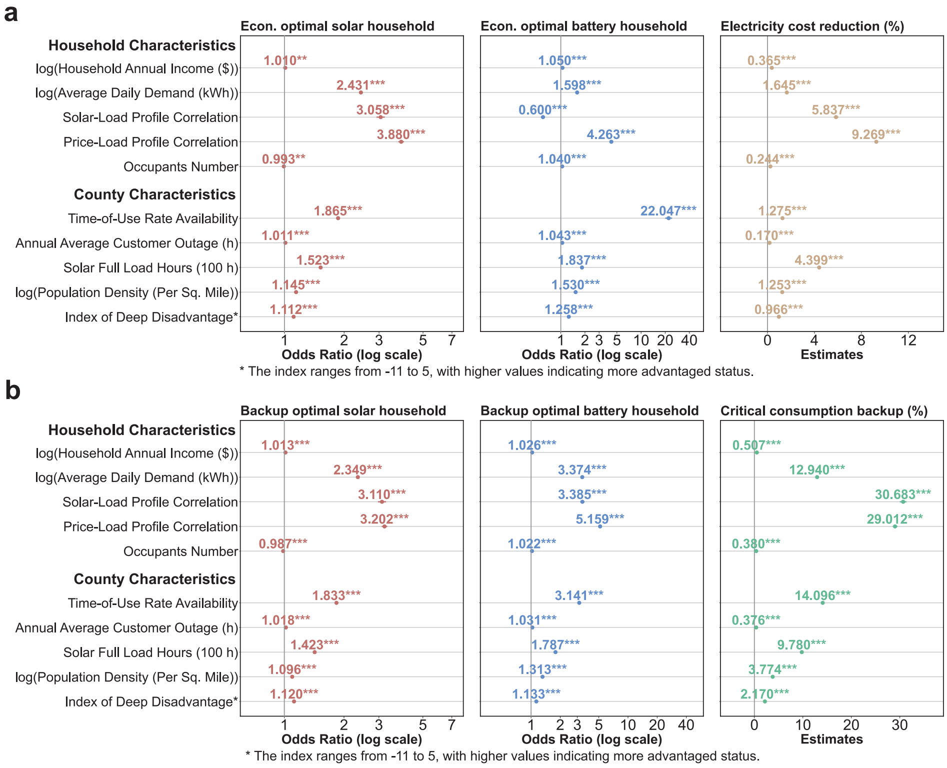

Fig. 7 Regression analysis of household solar-battery access under net billing tariffs using household and county characteristics. Household-level regression models (n=512,437) examining how various household and community characteristics shape access to solar-battery systems, measured through six key metrics spanning both economic and backup viability. Each plot displays separate regression model outputs for each of the following dependent variables: economically optimal solar households, economically optimal battery households, and electricity cost reduction (economic viability), as well as backup optimal solar households, backup optimal battery households, and critical consumption backup (backup viability). **a,** Logistic regression results for economically optimal solar/battery households and linear regression results for electricity cost reduction. **b,** Logistic regression results for backup optimal solar/battery households and linear regression results for critical consumption backup. Independent variables applied across all regression models include household-level traits (annual income, average daily electricity demand, solar-load profile correlation, price-load profile correlation, household occupancy) and community-level factors (time-of-use rate availability, annual average customer outage duration, solar full load hours, population density, and a deep disadvantage index). See Methods for details about variable construction. Given the large sample size, small differences can appear statistically significant; odds ratios near one (in logistic models) or estimates close to zero (in linear models) can therefore be interpreted as having negligible practical effects. Depending on the model specification, points represent either odds ratios or linear coefficient estimates with error bars indicating 95% confidence intervals; asterisks denote statistical significance (*$p<0.05$, **$p<0.01$, ***$p<0.001$).

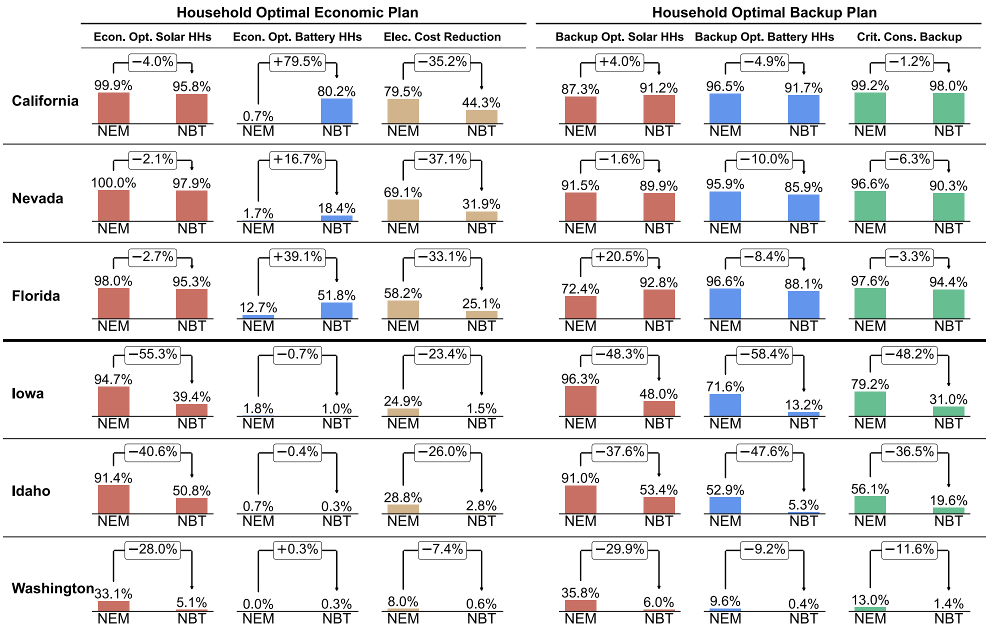

Fig. 8 State-level impacts of transitioning from net energy metering to net billing tariffs. Comparison of key solar-battery viability metrics before and after states shift from net energy metering to net billing tariffs, providing policymakers with data-driven insights for adjusting tariff structures across diverse regions. For each state, six metrics are shown: (1) proportion of economically optimal solar households, (2) proportion of economically optimal battery households, (3) state-level electricity bill-weighted average cost reduction, (4) proportion of backup optimal solar households, (5) proportion of backup optimal battery households, and (6) state-level essential energy needs-weighted average critical consumption backup. Distinct regional patterns emerge: the three states in the upper panel (California, Nevada, Florida) show only moderate impacts, maintaining overall viability while fostering battery adoption, whereas the three states in the lower panel (Iowa, Idaho, Washington) experience substantial declines in all metrics. Complete results for all states are available in Supplementary Table 11.

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